Pneumonia: Approach to a Case of Cough

Dr. Essam Kotb, Internal Medicine | Source: `11-Pneumonia Updated.pdf` (34 slides)

Legend: ⭐ = high-yield · *[added]* = not on slides. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes.

🔗 **Bridge from earlier respiratory lectures (same lecturer):**

- **COPD/asthma:** pneumonia is the classic trigger of exacerbations.
- **Lung abscess / bronchiectasis / CF:** aspiration and suppurative pneumonia.
- **Pleura: parapneumonic effusion → empyema** (Light's criteria; pH < 7.2 → drain).
- **Pulmonary vascular:** PE mimics pneumonia (pleuritic pain, fever, infiltrate).
- **Chest investigations:** consolidation signs and air bronchograms.

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1. Title / Objectives

Domain	Objective	Likely tested as
Knowledge	Define pneumonia; alveolar and interstitial inflammation	Definition
Knowledge	⭐ Classify by **acquisition, microbe, site**	CAP vs HAP vs VAP; lobar vs broncho vs interstitial
Knowledge	⭐ CAP organisms (**typical vs atypical**) and HAP organisms	"Fisherman with lobar consolidation → ?"
Knowledge	⭐ Typical vs atypical: onset, symptoms, **Gram stain**, imaging, **penicillin response**	Pulse–temperature dissociation
Knowledge	Risk factors for pneumonia and aspiration	
Knowledge	Complications; prevention (pneumococcal and influenza vaccine, smoking cessation)	Vaccinate before discharge
Skill	Read CXR/CT: **lobar consolidation, air bronchogram, reticulonodular infiltrates**	Image MCQ
Skill	⭐ **CURB-65** → outpatient / inpatient / ICU	Lecture case 2
Skill	⭐ Empiric regimen by risk and severity; **MDR risk → IV dual/triple therapy** for HAP	Regimen MCQ
Value	Professional conduct in training facilities	

Main subtopics: CAP · HAP · atypical pneumonia · aspiration pneumonia (definition, types, pathogenesis, presentation, diagnosis, management).

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2. Main Content (Lecture Order)

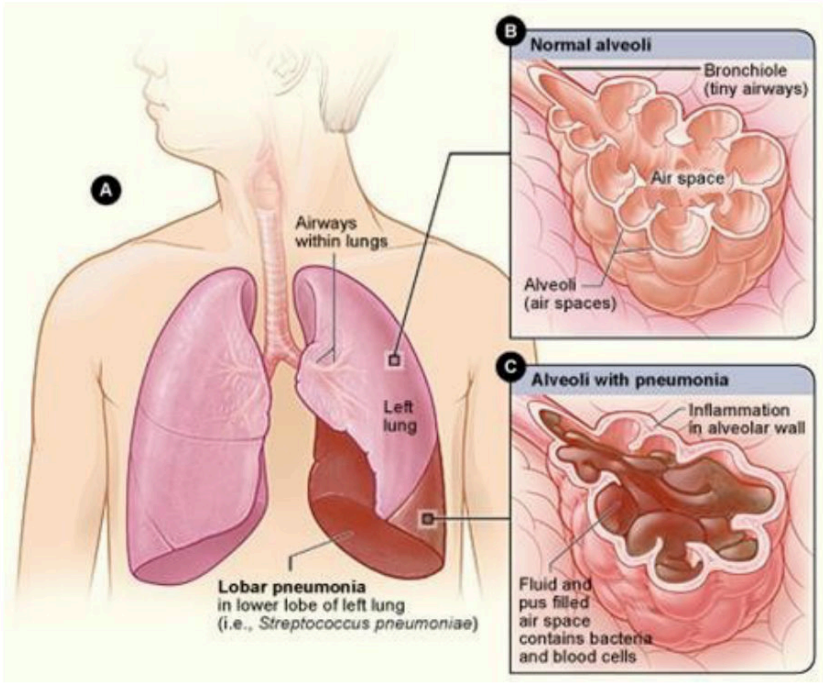
2.1 Definition & Transmission (Slide 5)

Pneumonia = respiratory infection with **inflammation of the alveolar space and/or the interstitial tissue** of the lungs.

- Most commonly from **aspiration of airborne pathogens** (mainly **bacteria**; also viruses and fungi).
- Can also result from **aspiration of stomach contents**.

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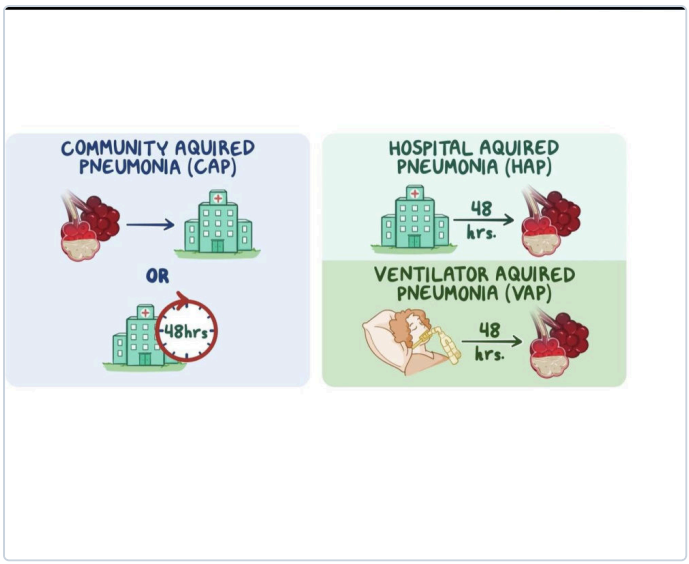
Slide 5: normal alveoli vs alveoli in pneumonia (lobar pneumonia)

2.2 Classification (Slides 6, 8) ⭐

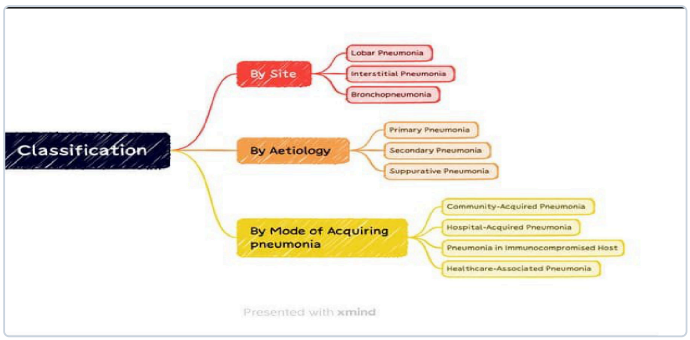
Basis	Types
Acquisition	**CAP** vs **HAP** vs **VAP** (slide 6 graphic: HAP and VAP both start **> 48 h** after admission/intubation)
Microbe	Bacterial vs viral vs fungal
Location	**Bronchopneumonia** vs **lobar** vs **interstitial (atypical)**
Aspiration	Separate category

Slide 8 mind-map:

- **By site:** lobar · interstitial · bronchopneumonia
- **By aetiology:** primary · secondary · suppurative
- **By mode of acquisition:** community-acquired · hospital-acquired · in the immunocompromised host · healthcare-associated



Slide 6: CAP vs HAP vs VAP (48 h rule)



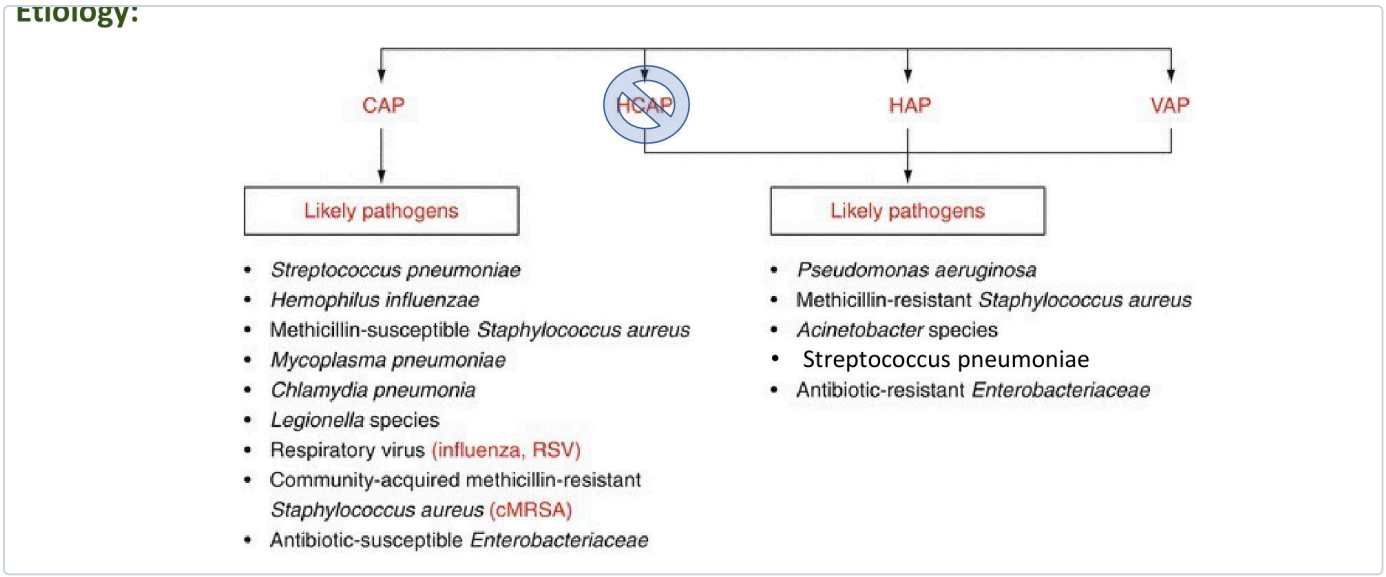
Slide 8: classification mind-map

[added] Lobar = a whole lobe, uniform (classically pneumococcus). Bronchopneumonia = patchy, centred on bronchioles, often bilateral and basal (S. aureus, H. influenzae, Gram-negatives, elderly/bedbound). Interstitial = alveolar walls/septa (viruses, Mycoplasma).

2.3 Aetiology by Setting (Slide 7) ⭐

CAP: likely pathogens	HAP / VAP: likely pathogens
Streptococcus pneumoniae	⭐ ***Pseudomonas aeruginosa***
Haemophilus influenzae	⭐ **MRSA**
MSSA	***Acinetobacter*** spp.
Mycoplasma pneumoniae	*S. pneumoniae*
Chlamydia pneumoniae	**Antibiotic-resistant Enterobacteriaceae**
Legionella	
Respiratory viruses (influenza, RSV)	
Community-acquired MRSA (cMRSA)	
Antibiotic-susceptible Enterobacteriaceae	

The slide shows HCAP (healthcare-associated) circled/struck out between CAP and HAP. [added] *It was removed in the 2016–2019 IDSA/ATS guidelines; those patients are now managed as CAP with MDR risk assessment.*



Slide 7: pathogens in CAP vs HAP/VAP

2.4 Community-Acquired: Typical vs Atypical (Slides 9–10) ⭐

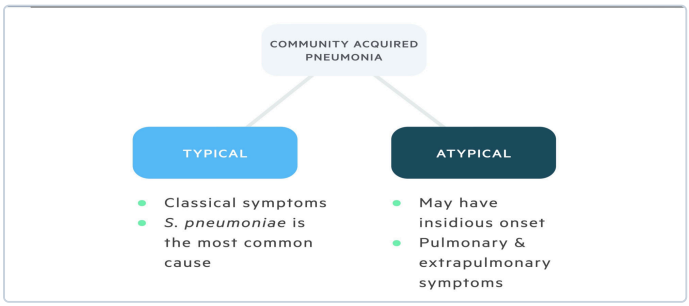
- **Typical:** classical symptoms; ***S. pneumoniae* is the most common cause**.
- **Atypical:** may have an **insidious onset**; **pulmonary and extrapulmonary** symptoms.

Typical CAP agents (slide 10):

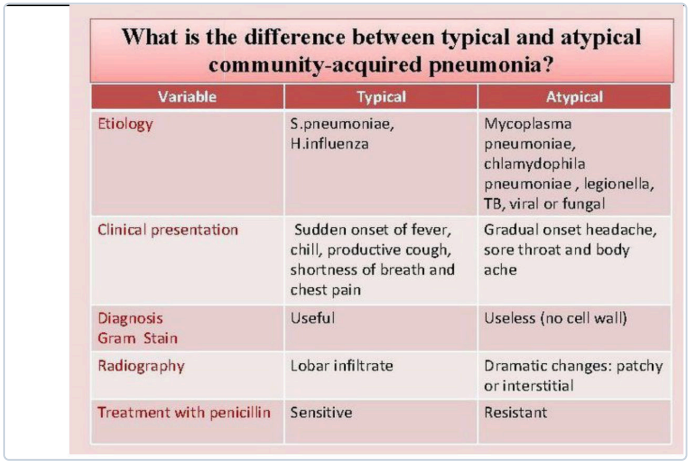
- ⭐ ***S. pneumoniae*: 60%**
- ***H. influenzae*: 15%**
- **Aerobic Gram-negative rods: 6–10%** (*Klebsiella* and other Enterobacteriaceae)
- ***S. aureus*: 2–10%**

Slide 10 table: typical vs atypical

Variable	Typical	Atypical
Aetiology	*S. pneumoniae*, *H. influenzae*	*Mycoplasma*, *Chlamydophila pneumoniae*, *Legionella*, **TB, viral or fungal**
Presentation	**Sudden** fever, **chills**, **productive cough**, dyspnoea, chest pain	**Gradual** headache, **sore throat, body ache**
Gram stain	**Useful**	⭐ **Useless (no cell wall)**
Radiography	**Lobar infiltrate**	**Dramatic changes: patchy or interstitial**
Penicillin	**Sensitive**	⭐ **Resistant**



Slide 9: typical vs atypical CAP

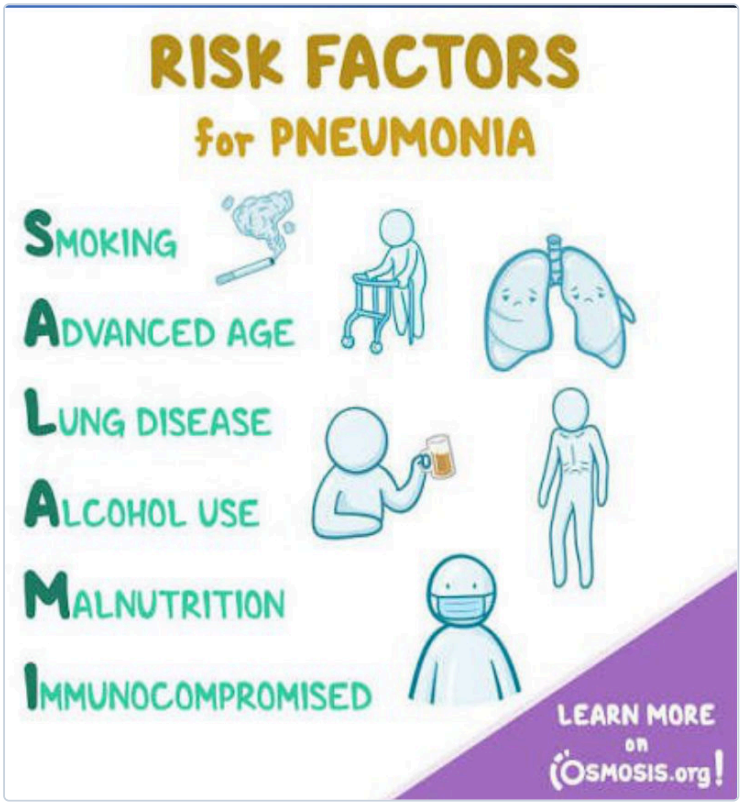


Slide 10: typical vs atypical table

2.5 Risk Factors (Slide 11)

- **Old age, immobility** of any cause
- **Chronic disease:** - **cardiopulmonary** disease (asthma, **COPD**, **HF**) - **airway abnormalities** (**bronchiectasis**, space-occupying lesions, **CF**)
- **Immunosuppression**: **HIV**, **DM**, immunosuppressive therapy
- ⭐ **Impaired airway protection**: **altered consciousness** (stroke, seizure, anaesthesia, drugs, **alcohol**)
- **Environment**: crowded living; **contaminated water systems** (hotels, cruise ships → *Legionella*)
- **Surgery**

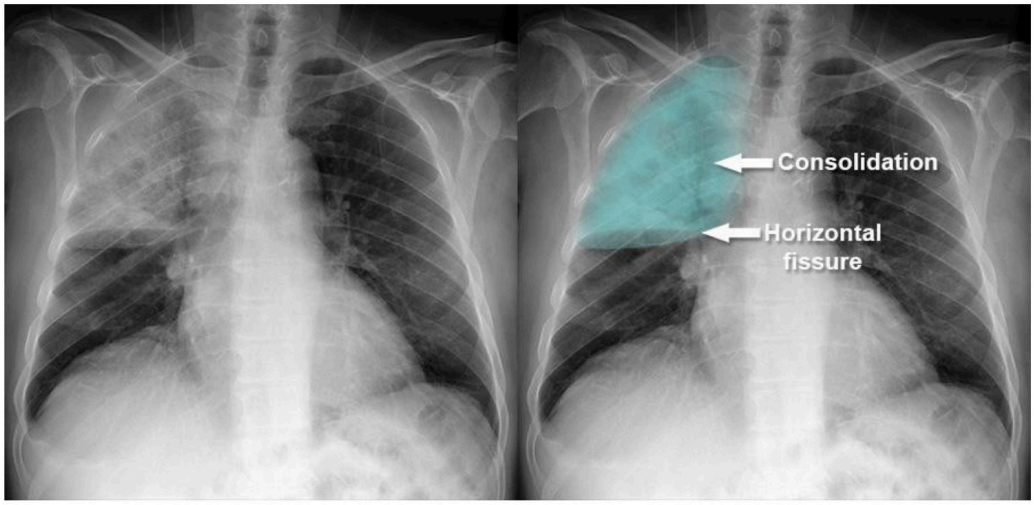
> **Mnemonic (slide 11 Osmosis graphic): "SALAMI"** = **S**moking · **A**dvanced age · **L**ung disease · **A**lcohol use · **M**alnutrition · **I**mmunocompromised.



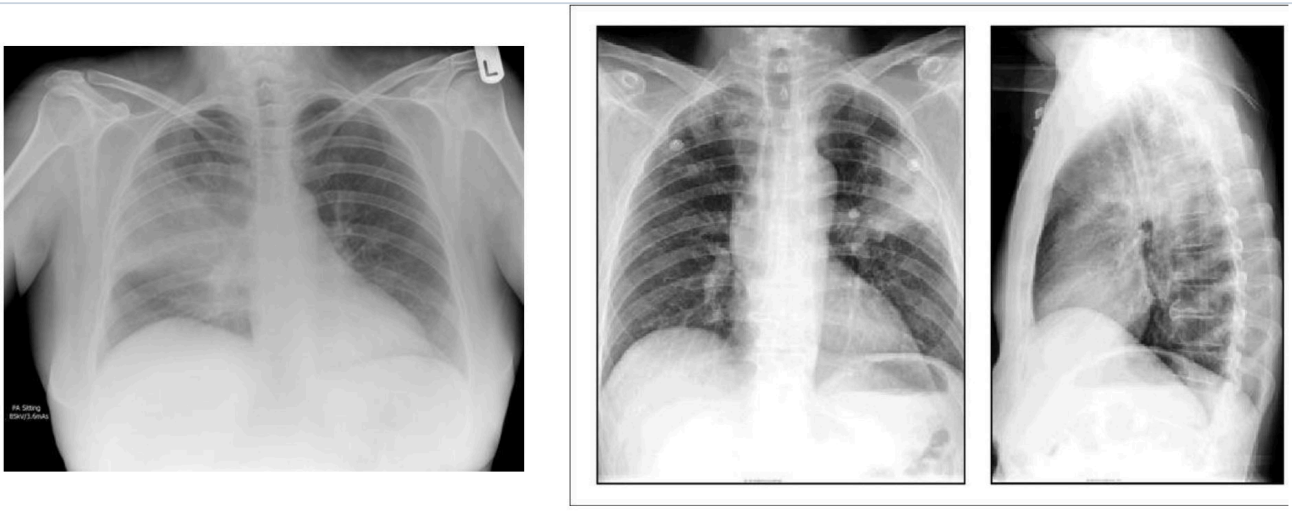
Slide 11: risk factors for pneumonia (SALAMI)

2.6 Typical CAP: Clinical Picture (Slides 12–14) ⭐

- **Symptoms:** - **acute** fever with **shaking chills (rigors)** - **productive cough** with thick **purulent** sputum (*[added]* rusty in pneumococcus; currant-jelly in *Klebsiella*) - **pleuritic chest pain** - dyspnoea
- **Signs:** - tachycardia, tachypnoea - ⭐ consolidation signs: **late inspiratory crackles, bronchial breath sounds, ↑ tactile and vocal fremitus, dullness** - **pleural friction rub** (pleurisy)
- **CXR:** ⭐ **lobar consolidation**. **Multilobar = very serious illness**.



Slide 13: right upper lobe consolidation bordered by the horizontal fissure

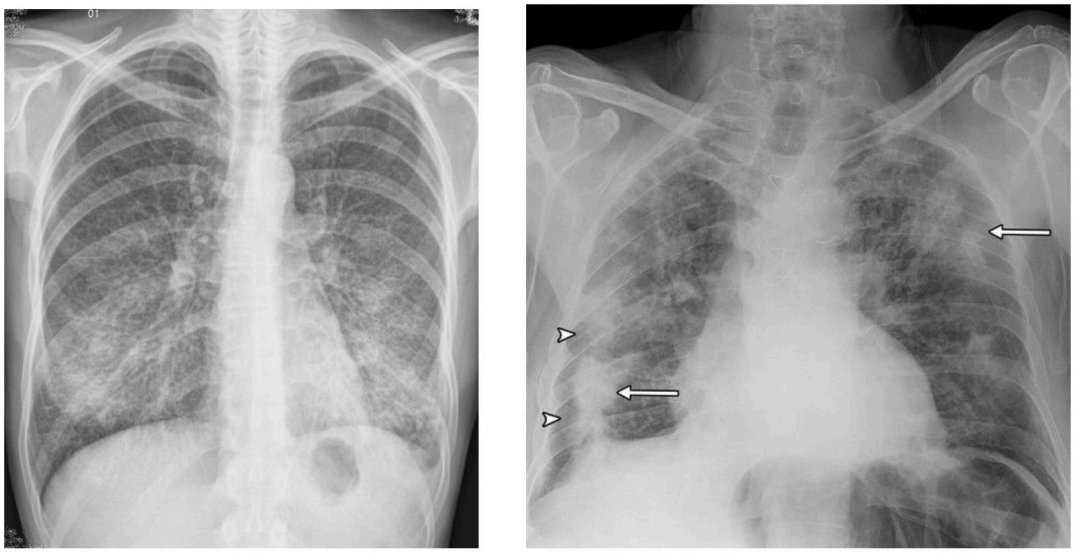


Slide 14: lobar pneumonia, PA and lateral

[added] Slide 13: RUL consolidation with a sharp inferior border along the horizontal fissure = classic lobar pneumonia of the right upper lobe.

2.7 Atypical CAP: "Walking Pneumonia" (Slides 15–18) ⭐

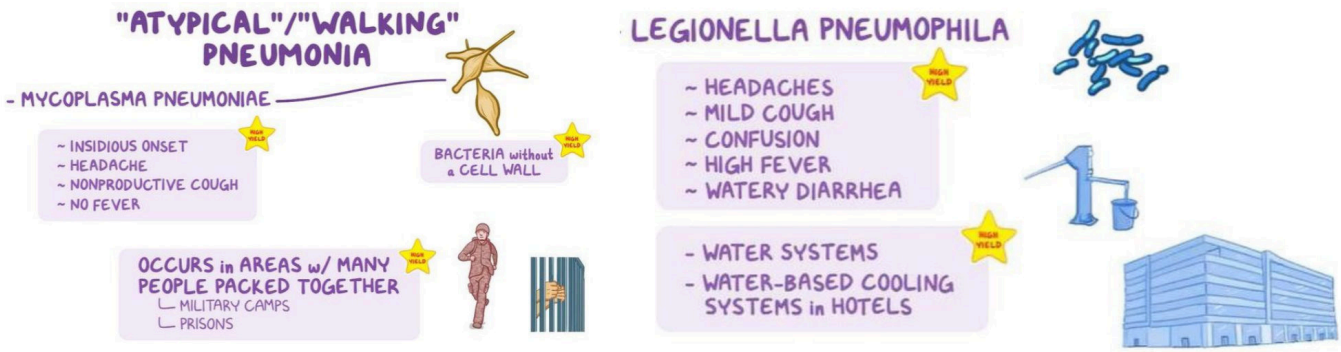
- **Organisms:** - ⭐ ***Mycoplasma pneumoniae* (most common)** - *Chlamydia pneumoniae* - ***C. psittaci*** - ***Coxiella burnetii*** (Q fever) - ***Legionella*** - **viruses**: influenza A/B, adenovirus, parainfluenza, RSV
- ⭐ **Quick hit (slide 15):** "atypical" = organisms **not visible on Gram stain** and **not culturable on standard blood agar**.
- **Symptoms:** - **insidious** onset: headache, sore throat, fatigue, myalgia - ⭐ **dry cough** - fever, but **chills uncommon**
- **Signs:** - ⭐ **pulse–temperature dissociation**: a normal pulse with high fever - wheeze, rhonchi, crackles
- **CXR:** ⭐ **diffuse reticulonodular infiltrates**; absent or minimal consolidation.



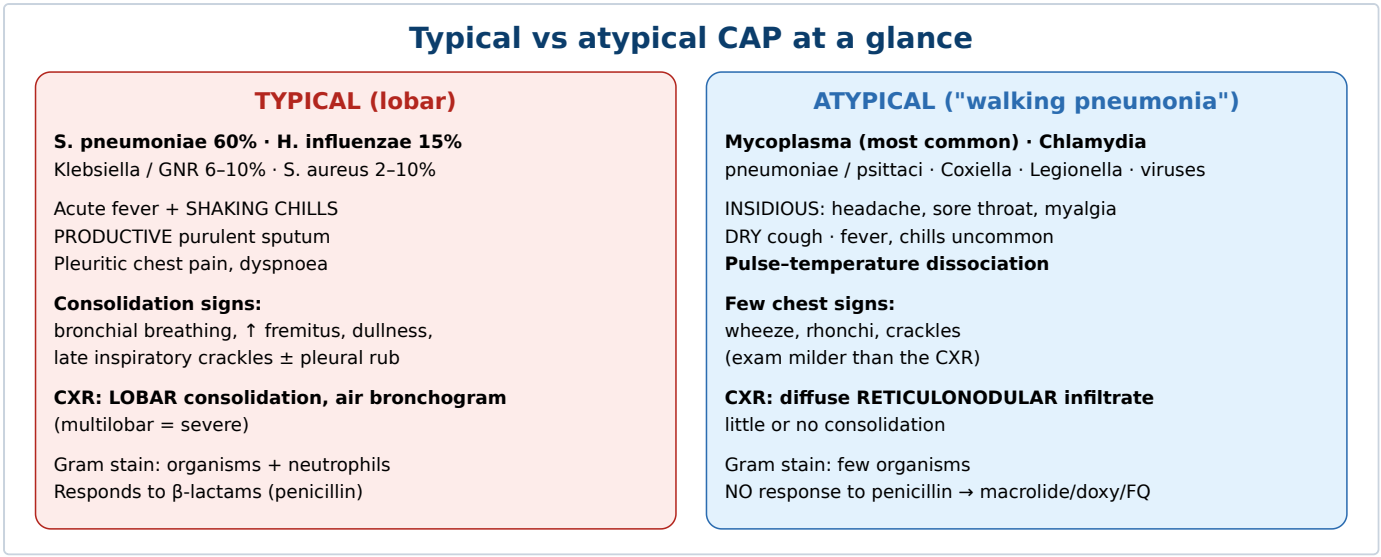
Slide 17: atypical pneumonia, diffuse interstitial/reticulonodular change

Slide 18 (Osmosis):

- **Mycoplasma: mildly ill, insidious onset**, headache, **non-productive cough**, low fever. **Bacteria without a cell wall.** Occurs where **many people are packed together** (military camps, prisons).
- ⭐ ***Legionella***: headaches, mild cough, **confusion**, **high fever**, **watery diarrhoea**. Source: **water systems**, **water-based cooling systems in hotels**.



Slide 18: Mycoplasma vs Legionella



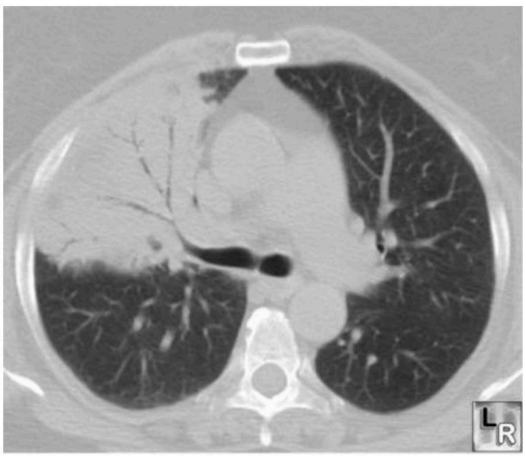
Added diagram: typical vs atypical CAP at a glance

[added] Organism clues:

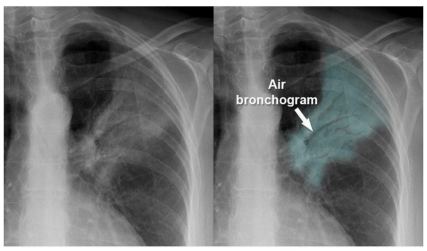
Clue	Think
Young adult in a dormitory or barracks; **cold agglutinins**, bullous myringitis, erythema multiforme	*Mycoplasma*
Hyponatraemia, diarrhoea, confusion, ↑ LFTs; hotel, cruise, air-conditioning	*Legionella* (**urinary antigen**)
Birds	*C. psittaci*
Farm animals	*Coxiella*
Alcoholic; **currant-jelly sputum**	*Klebsiella*
After influenza; cavitation	*S. aureus*
COPD smoker	*H. influenzae*, *Moraxella*
Aspiration; foul sputum; right lower lobe	Anaerobes

2.8 Diagnostics (Slides 19–21) ⭐

- **CXR (PA + lateral)**
- **CBC**; **↑ CRP, ↑ ESR**; ⭐ **↑ procalcitonin**
- **Chest CT** (without contrast)
- **Sputum Gram stain and culture**; **blood cultures**
- **Flu swab**; **influenza and COVID-19 testing**
- **↓ SpO_2**
- ⭐ **Urinary antigen for *Legionella*** (when suspected)



Slide 20: CT chest, consolidation with air bronchograms



Slide 21: air bronchogram

[added] Air bronchogram = air-filled bronchi seen against fluid-filled alveoli. It is the hallmark of alveolar consolidation, not effusion or collapse.

🔁 **Recap: Recognising pneumonia.**

- **Definition:** alveolar and/or interstitial infection. Classify by acquisition (CAP/HAP/VAP; > 48 h), microbe and site.
- **CAP organisms:** pneumococcus 60% > H. influenzae 15% > GNR > S. aureus. **HAP organisms:** Pseudomonas, MRSA, Acinetobacter, resistant Enterobacteriaceae.
- **Typical:** acute, rigors, purulent sputum, consolidation signs, lobar CXR, Gram stain useful, penicillin-sensitive.
- **Atypical:** Mycoplasma commonest; insidious, dry cough, pulse–temperature dissociation, reticulonodular CXR, Gram stain useless, penicillin-resistant. Legionella = confusion + diarrhoea + water systems.
- **Tests:** CXR, inflammatory markers, procalcitonin, cultures, flu/COVID swab, Legionella urinary antigen.

🔗 **Link:** Once diagnosed, decide where to treat (CURB-65), then choose the drugs.

▶️ **Up next: Management.**

2.9 Where to Treat: CURB-65 (Slide 22) ⭐

Symptom	Points
Confusion	1
Urea: **BUN > 19 mg/dL (> 7 mmol/L)**	1
Respiratory rate **≥ 30/min**	1
BP: **systolic < 90 or diastolic ≤ 60 mmHg**	1
Age **≥ 65**	1

⭐ Score	Mortality	Disposition
0–1	**1.5%**	**Outpatient care**
2	**9.2%**	**Hospitalisation is indicated**

⭐ Score	Mortality	Disposition
≥ 3	**22%**	**Consider ICU admission**

Table 1. CURB-65 Scoring[46]

Symptom	Points
Confusion	1
Urea: BUN >19 mg/dL (>7 mmol/L)	1
Respiratory rate ≥30 breaths/min	1
Systolic BP <90 mm Hg or diastolic BP ≤60 mm Hg	1
Age ≥65 years	1
Total	_____

Score	Risk	Disposition
0 or 1	1.5% mortality	Outpatient care
2	9.2% mortality	Hospitalization is indicated.
≥3	22% mortality	Consider ICU admission.

Abbreviations: BP, blood pressure; BUN, blood urea nitrogen; ICU, intensive care unit.

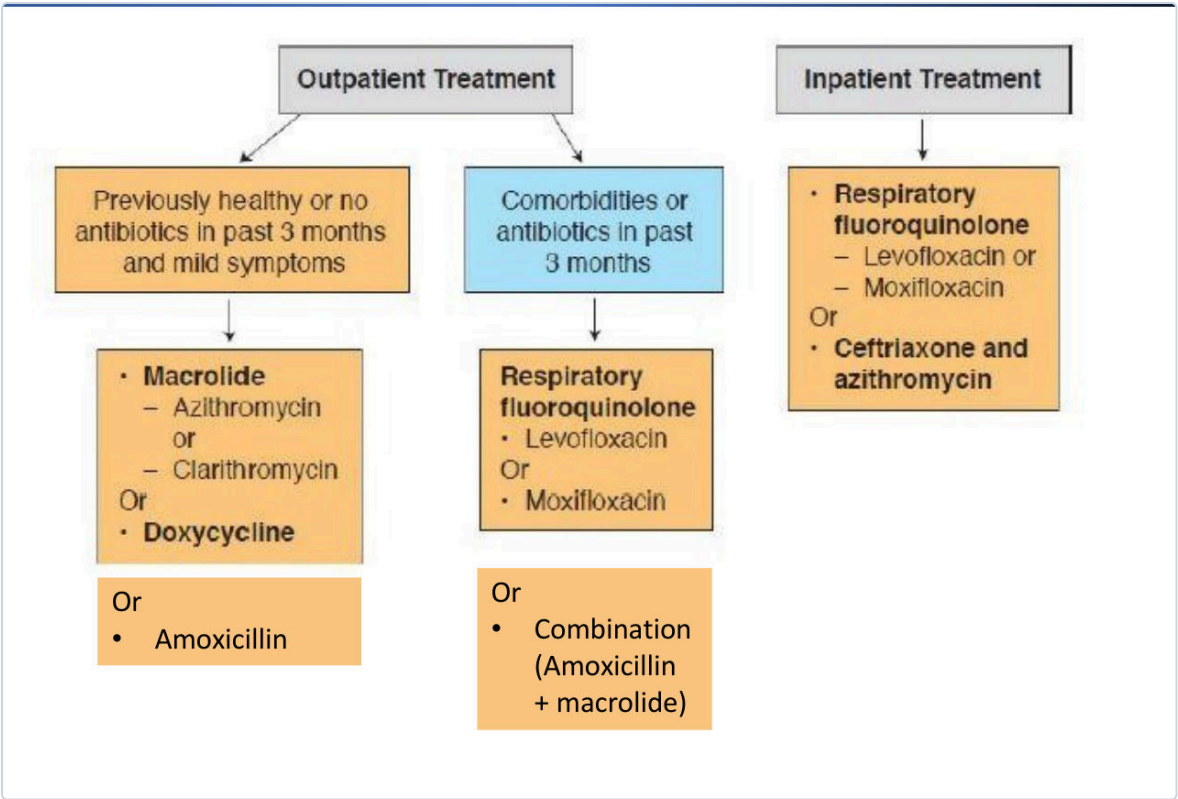
Slide 22: CURB-65 scoring and disposition

2.10 Empiric Management (Slide 23) ⭐

General:

- **Empiric antibiotics** by severity and risk factors (e.g., VAP vs CAP)
- **Antivirals** for influenza and COVID-19 as indicated
- **Identify and manage sepsis**; **fluid resuscitation** for septic shock
- **Respiratory support** if needed; **O_2** for hypoxia

⭐ Setting	Regimen (slide 23)
Outpatient: previously healthy, no antibiotics in the past 3 months, mild symptoms	**Macrolide** (azithromycin or clarithromycin) **OR doxycycline OR amoxicillin**
Outpatient: comorbidities, or antibiotics in the past 3 months	**Respiratory fluoroquinolone** (levofloxacin or moxifloxacin) **OR combination (amoxicillin + macrolide)**
Inpatient	**Respiratory FQ** (levofloxacin or moxifloxacin) **OR ceftriaxone + azithromycin**



Slide 23: empiric outpatient vs inpatient regimens

[added] ICU/severe CAP (IDSA/ATS 2019): β-lactam + macrolide or β-lactam + FQ. Add MRSA or Pseudomonas cover only with risk factors. ICU criteria: 1 major (vasopressors, mechanical ventilation) or ≥ 3 minor (RR ≥ 30, P/F ≤ 250, multilobar, confusion, BUN ≥ 20, WBC < 4000, platelets < 100 000, T < 36 °C, hypotension needing aggressive fluids).

2.11 Targeted CAP Treatment (Slide 24): Table 94.1

Agent	Preferred	Alternative
S. pneumoniae	**Penicillin**	Cephalosporin, moxifloxacin, levofloxacin
H. influenzae	**Cefuroxime, ceftriaxone**	
Mycoplasma	**Macrolide**	Moxifloxacin, levofloxacin
Legionella	**Macrolide or quinolone**	
***S. aureus*, methicillin-susceptible**	**Nafcillin**	Cephalosporin
***S. aureus*, methicillin-resistant**	⭐ **Vancomycin (IV)**	Doxycycline or TMP/SMX (oral)
Moraxella catarrhalis	**Amox-clav, cefuroxime, ceftriaxone, TMP/SMX**	

TABLE 94.1 Treatment of Pneumonia by Specific Etiologic Agent

Etiologic Agent	Preferred Antimicrobial	Alternative Antimicrobial
Streptococcus pneumoniae	Penicillin	Cephalosporin, moxifloxacin, levofloxacin
Haemophilus influenzae	Cefuroxime, ceftriaxone	
Mycoplasma pneumoniae	Macrolide	Moxifloxacin, levofloxacin
Legionella	Macrolide or quinolone	
Staphylococcus aureus		
Methicillin-susceptible	Nafcillin	Cephalosporin
Methicillin-resistant	Vancomycin (IV)	Doxycycline or TMP/SMX (oral)
Moraxella catarrhalis	Amoxicillin/clavulanate, cefuroxime, ceftriaxone, TMP/SMX	

TMP/SMX, Trimethoprim/sulfamethoxazole

Slide 24: treatment by aetiologic agent

2.12 Prevention (Slide 25) ⭐

- ⭐ **Pneumococcal vaccine: give to ALL patients with CAP before discharge** unless already vaccinated.
- **Indications (US table):** - **all adults ≥ 65** - **adults 19–64** with: - **predisposing conditions**: alcohol use disorder; chronic heart, lung or liver disease; **DM**; **sickle cell / haemoglobinopathy**; **current smoking** - **↑ meningitis risk**: **CSF leak**, **cochlear implant** - **immunocompromise**: congenital or acquired immunodeficiency, generalised malignancy, **HIV**, iatrogenic immunosuppression, Hodgkin's, leukaemia, lymphoma, **myeloma**, **solid organ transplant**, **CKD / nephrotic syndrome**, ⭐ **functional or anatomic asplenia** - **history of invasive pneumococcal disease**
- **Influenza vaccine.**
- **Smoking cessation.**

Indications for pneumococcal vaccination in adults in the United States

All adults ≥65 years of age

Adults 19 to 64 years of age with any of the following:

- Predisposing medical conditions:
  - Alcohol use disorder
  - Chronic heart disease*
  - Chronic lung disease¶
  - Chronic liver disease
  - Diabetes mellitus
  - Sickle cell disease or other hemoglobinopathies
  - Current cigarette smoking
- Increased risk of meningitis:
  - Cerebrospinal fluid leak
  - Cochlear implant
- Immunocompromising conditions and other conditions associated with altered immunocompetence△:
  - Congenital or acquired immunodeficiency◇
  - Generalized active malignancy
  - Human immunodeficiency virus infection§
  - Iatrogenic immunosuppression¥
  - Hodgkin disease
  - Leukemia
  - Lymphoma
  - Multiple myeloma
  - Solid organ transplant
  - Chronic kidney disease‡ and nephrotic syndrome
  - Functional or anatomic asplenia
- History of invasive pneumococcal disease†

Slide 25: indications for pneumococcal vaccination

🔁 **Recap: CAP management.**

- **CURB-65:** confusion, BUN > 19, RR ≥ 30, SBP < 90/DBP ≤ 60, age ≥ 65. 0–1 → home (1.5%); 2 → admit (9.2%); ≥ 3 → consider ICU (22%).
- **Outpatient, healthy:** macrolide or doxycycline or amoxicillin. **Comorbid or recent antibiotics:** respiratory FQ or amoxicillin + macrolide. **Inpatient:** respiratory FQ or ceftriaxone + azithromycin.
- **Targeted:** pneumococcus → penicillin; Mycoplasma → macrolide; Legionella → macrolide or FQ; MRSA → vancomycin.
- **Prevention:** pneumococcal vaccine before discharge, flu vaccine, stop smoking.

🔗 **Link:** In hospital, the organisms change, and so does the empiric cover.

▶️ **Up next: Hospital-acquired pneumonia.**

2.13 Hospital-Acquired Pneumonia (Slides 26–27) ⭐

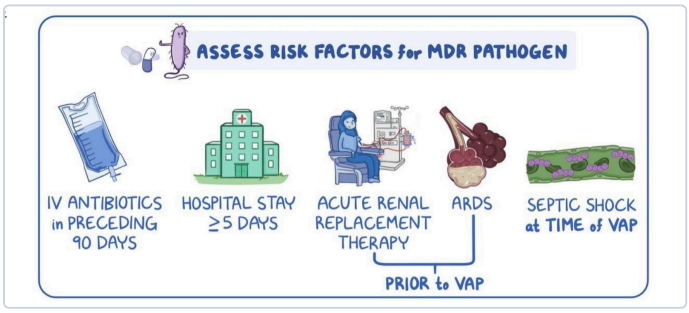
- **HAP:** onset **> 48 h after admission** with **no evidence of disease at admission**.
- **VAP: > 48 h after intubation** (slide 6).

Step 1: assess risk factors for MDR pathogens (slide 26)

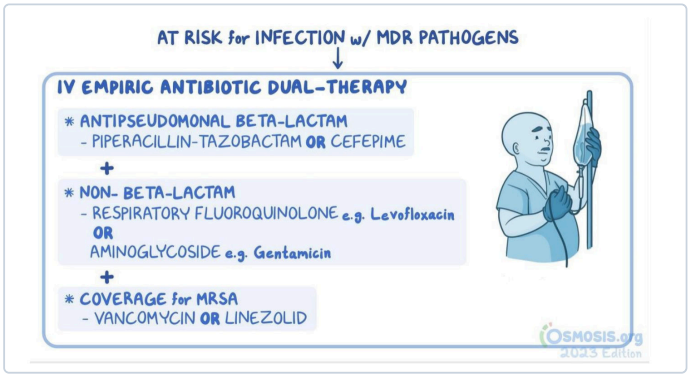
- **IV antibiotics in the preceding 90 days**
- **Hospital stay ≥ 5 days**
- **Acute renal replacement therapy** (prior to VAP)
- **ARDS** (prior to VAP)
- **Septic shock at the time of VAP**

Step 2: MDR risk → IV empiric DUAL therapy (slide 27)

1. **Anti-pseudomonal β-lactam**: **piperacillin-tazobactam OR cefepime**
2. **+ Non-β-lactam**: **respiratory FQ (levofloxacin) OR aminoglycoside (gentamicin)**
3. **+ MRSA cover**: **vancomycin OR linezolid** (→ "triple")



Slide 26: MDR risk factors



Slide 27: IV empiric dual therapy ± MRSA cover

[added] Without MDR risk (and with low mortality risk), a single anti-pseudomonal agent is enough (pip-tazo, cefepime, levofloxacin or a carbapenem). Never give two β-lactams together. De-escalate once cultures return.

2.14 Aspiration Pneumonia *(listed as a subtopic; no dedicated slide)*

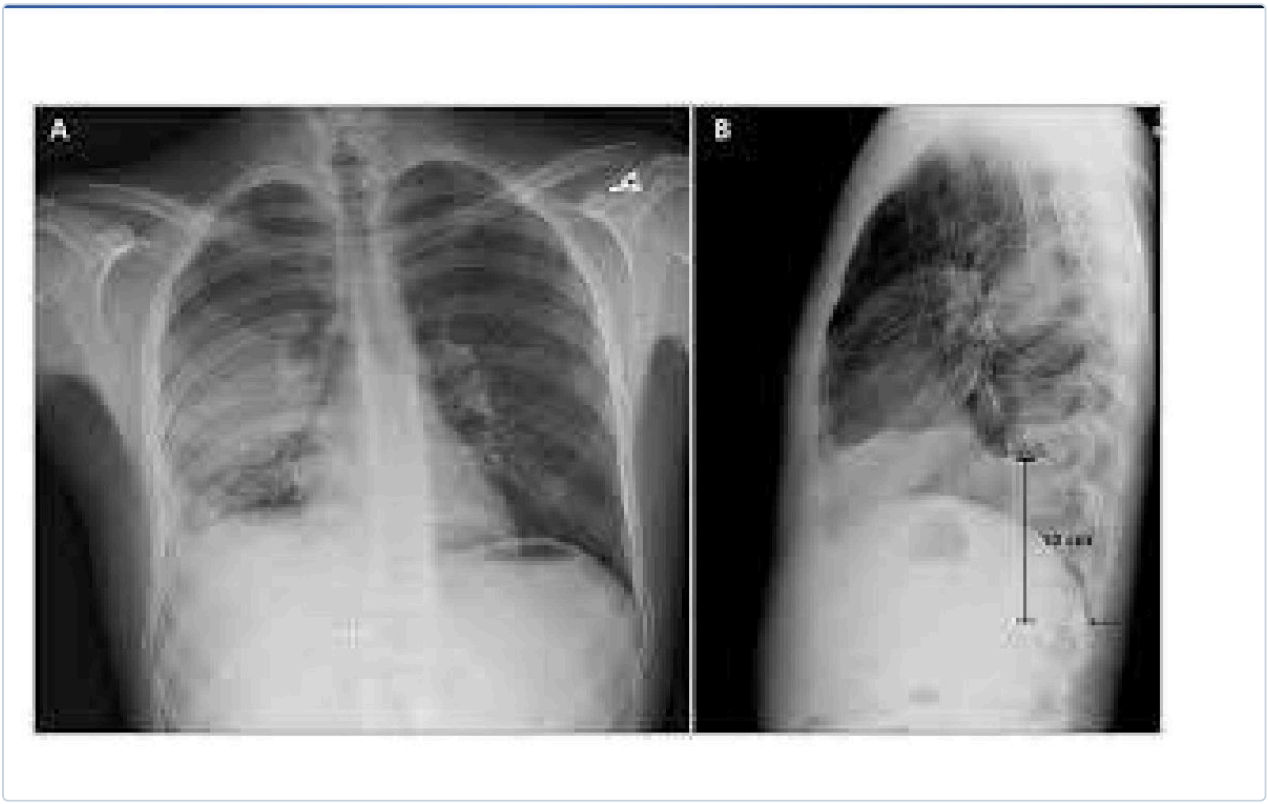
[added]:

- **Risk:** ↓ consciousness (stroke, seizure, alcohol, sedation), dysphagia, poor dentition.

- **Site: right lower lobe** when upright; posterior upper lobe / superior lower-lobe segment when supine.
- **Aspiration pneumonitis** (chemical, Mendelson) vs **aspiration pneumonia** (anaerobes and oral flora).
- **Treatment:** amp-sulbactam or clindamycin. Risk of **abscess and empyema**.

2.15 Complications (Slide 28)

- **Parapneumonic pleuritis**
- **Parapneumonic pleural effusion** → ⭐ **pleural empyema**
- **Lung abscess**
- **ARDS** → **respiratory failure**
- **Sepsis**

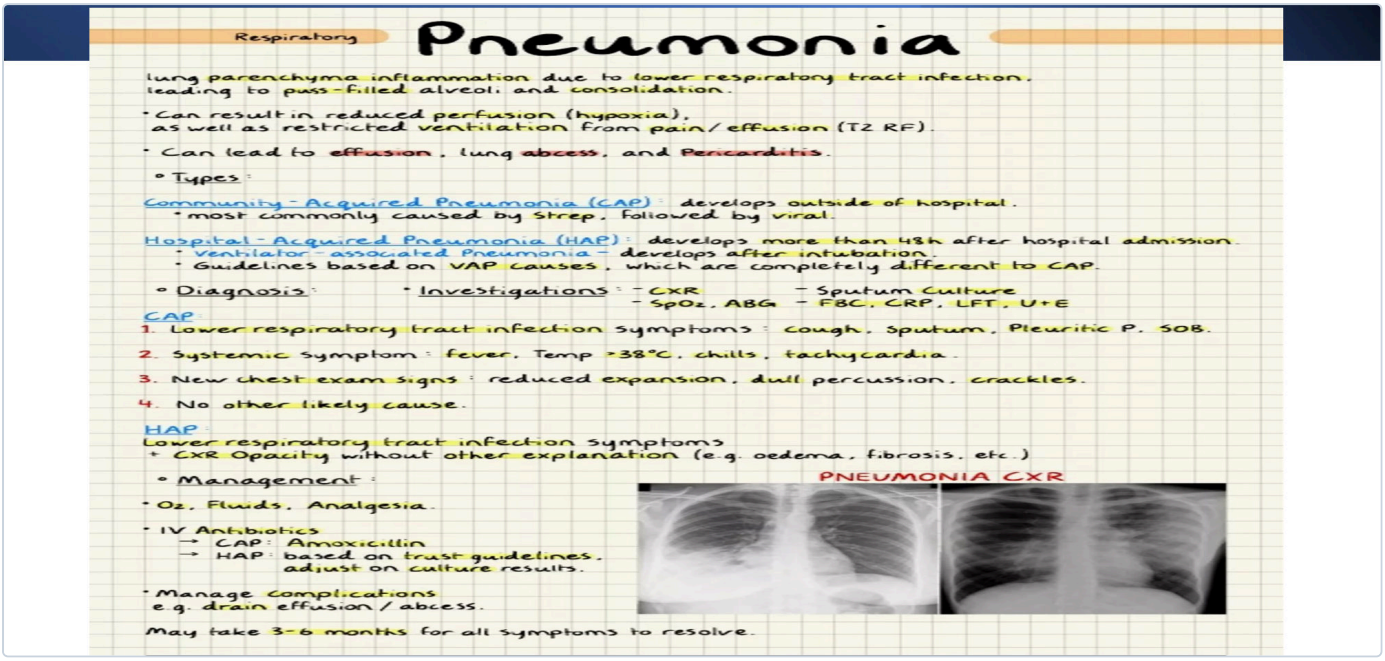


Slide 28: right lower zone pneumonia with parapneumonic effusion (PA + lateral)

2.16 Lecturer's Summary (Slide 29)

- **Definition:** lung parenchymal inflammation from lower respiratory infection → **pus-filled alveoli and consolidation**.
- Causes **↓ perfusion (hypoxia)** and **restricted ventilation** from pain or effusion (**type 2 respiratory failure**).
- **Can lead to effusion, lung abscess and pericarditis.**
- **CAP:** develops outside hospital; most commonly **Strep, then viral**.
- **HAP:** > 48 h after admission. **VAP** after intubation. Guidelines are based on VAP causes, which are **completely different** from CAP.
- **Investigations:** CXR, SpO_2, ABG, sputum culture, FBC, CRP, LFT, U&E.
- **CAP diagnosis (all four):** 1. **LRTI symptoms**: cough, sputum, pleuritic pain, dyspnoea 2. **Systemic**: fever > 38 °C, chills, tachycardia 3. **New chest signs**: ↓ expansion, dull percussion, crackles 4. **No other likely cause**
- **HAP diagnosis:** LRTI symptoms **+ a new CXR opacity** without another explanation (oedema, fibrosis…).

- **Management:** - **O_2, fluids, analgesia** - IV antibiotics: **CAP → amoxicillin**; **HAP → by trust guidelines**, then adjust to cultures - manage complications (drain effusion or abscess)
- ⭐ **Symptoms may take 3–6 months to fully resolve.**



Slide 29: lecturer's summary sheet

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3. High-Yield Comparison Tables

Table A: Typical vs Atypical CAP

	Typical	Atypical
Organisms	**S. pneumoniae 60%**, H. influenzae 15%, GNR 6–10%, S. aureus 2–10%	**Mycoplasma (commonest)**, C. pneumoniae, C. psittaci, Coxiella, Legionella, viruses
Onset	**Sudden**	**Gradual / insidious**
Fever / chills	High fever + **shaking chills**	Fever, **chills uncommon**
Cough	**Productive, purulent**	**Dry**
Pulse	Tachycardia	**Pulse–temperature dissociation**
Exam	**Consolidation signs**	Wheeze, rhonchi, crackles
CXR	**Lobar**	**Patchy/interstitial, reticulonodular** ("dramatic changes")
Gram stain	**Useful**	**Useless (no cell wall / not on blood agar)**
Penicillin	**Sensitive**	**Resistant**

Table B: CAP vs HAP vs VAP

	CAP	HAP	VAP
Timing	Community	**> 48 h after admission**	**> 48 h after intubation**
Organisms	Pneumococcus, H. influenzae, MSSA, atypicals, viruses, cMRSA	**Pseudomonas, MRSA, Acinetobacter**, resistant Enterobacteriaceae	Same as HAP

	CAP	HAP	VAP
Empiric	By CURB-65 and comorbidity (table 2.10)	**MDR risk → pip-tazo/cefepime + FQ/aminoglycoside ± vanc/linezolid**	Same; MDR risk includes RRT/ARDS before VAP, septic shock at VAP

Table C: Empiric CAP Regimens (Slide 23)

Patient	Drug
Healthy outpatient, no recent antibiotics	**Macrolide / doxycycline / amoxicillin**
Comorbid or antibiotics in past 3 months	**Resp. FQ** or **amoxicillin + macrolide**
Inpatient	**Resp. FQ** or **ceftriaxone + azithromycin**

Table D: Consolidation vs Effusion on Exam *[added]*

Sign	Consolidation	Effusion
Percussion	**Dull**	**Stony dull**
Fremitus	**↑**	**↓**
Breath sounds	**Bronchial**	**↓ / absent**
Crackles	Present	Absent

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4. Rules, Exceptions, Patterns

Rule	Exception / nuance
Pneumonia → bacteria	Viruses (influenza, COVID-19) are common: test and treat with antivirals
Fever → tachycardia	**Pulse–temperature dissociation** in atypicals; β-blockers also blunt the pulse
Penicillin for CAP	**Atypicals are resistant** (no cell wall): macrolide, doxycycline or FQ
Gram stain identifies the organism	**Useless in atypicals**
One lobe = typical	**Multilobar = very serious illness**
Vaccinate later	⭐ **Pneumococcal vaccine before discharge**
HAP = any hospital pneumonia	Only if onset > 48 h after admission and absent at admission
Single agent for HAP	**MDR risk → dual anti-pseudomonal + MRSA cover**
Pneumonia resolves in days	**Symptoms may take 3–6 months** to resolve fully (slide 29)

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5. Clinical Correlations

- **Elderly diabetic with rigors, rusty sputum, RUL consolidation, CURB-65 2** → admit → **ceftriaxone + azithromycin** (or resp. FQ).
- **Student in a dormitory with dry cough, headache, sore throat, normal pulse, patchy infiltrates** → *Mycoplasma* → macrolide or doxycycline.
- **Hotel guest with high fever, confusion, watery diarrhoea, low Na** → *Legionella* → **urinary antigen**; macrolide or FQ.
- **Post-stroke patient with right lower lobe pneumonia and foul sputum** → aspiration → amp-sulbactam.

- **Day 6 in hospital after IV antibiotics, new fever and infiltrate** → **HAP with MDR risk** → **pip-tazo + levofloxacin + vancomycin**.
- **Fever persisting on antibiotics, enlarging effusion** → tap for **empyema** (pH < 7.2 → drain).
- **Asplenic or sickle cell patient** → pneumococcal vaccination indicated.

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6. Rapid Revision (Last-Day)

1. Pneumonia = alveolar and/or interstitial inflammation, mostly from aspirated airborne pathogens.
2. Classify: **CAP / HAP (> 48 h) / VAP (> 48 h after intubation)**; bacterial/viral/fungal; **lobar / broncho / interstitial**; aspiration.
3. CAP: **S. pneumoniae 60%**, H. influenzae 15%, GNR 6–10%, S. aureus 2–10%. **HAP: Pseudomonas, MRSA, Acinetobacter**, resistant Enterobacteriaceae.
4. Risk factors (**SALAMI**): age/immobility, COPD/asthma/HF, bronchiectasis/CF, HIV/DM/immunosuppression, ↓ consciousness, crowding, contaminated water, surgery.
5. Typical: sudden, rigors, purulent sputum, pleuritic pain; **dull, ↑ fremitus, bronchial breathing, crackles**; **lobar** (multilobar = severe); Gram stain useful; penicillin-sensitive.
6. Atypical: **Mycoplasma** commonest; insidious, **dry cough**, chills uncommon, **pulse–temperature dissociation**, **reticulonodular**; Gram stain **useless**; penicillin-**resistant**. Legionella: confusion, high fever, **watery diarrhoea**, water systems.
7. Tests: CXR, CBC, CRP/ESR, **procalcitonin**, CT, sputum and blood cultures, flu/COVID swab, SpO_2, **Legionella urinary antigen**.
8. **CURB-65:** Confusion, **BUN > 19**, RR ≥ 30, SBP < 90 / DBP ≤ 60, age ≥ 65. **0–1 home (1.5%) · 2 admit (9.2%) · ≥ 3 consider ICU (22%)**.
9. **Outpatient healthy:** macrolide / doxycycline / amoxicillin. **Comorbid or antibiotics in past 3 months:** resp. FQ or amoxicillin + macrolide. **Inpatient:** resp. FQ or **ceftriaxone + azithromycin**.
10. Targeted: pneumococcus → **penicillin**; H. influenzae → cefuroxime/ceftriaxone; Mycoplasma → **macrolide**; Legionella → macrolide or FQ; MSSA → **nafcillin**; MRSA → **vancomycin**; Moraxella → amox-clav, cefuroxime, ceftriaxone, TMP/SMX.
11. Also: antivirals as indicated, sepsis care, fluids, O_2, respiratory support.
12. **Pneumococcal vaccine before discharge**; flu vaccine; stop smoking. Vaccinate all ≥ 65 and at-risk 19–64 (asplenia, CSF leak, cochlear implant, HIV, CKD, smokers…).
13. **HAP MDR risk:** IV antibiotics in 90 days, stay ≥ 5 days, RRT or ARDS before VAP, septic shock at VAP → **pip-tazo/cefepime + levofloxacin/gentamicin + vancomycin/linezolid**.
14. Complications: pleuritis, **parapneumonic effusion/empyema**, abscess, **ARDS**, respiratory failure, **sepsis** (and pericarditis, per the summary).
15. Recovery may take **3–6 months**.

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7. Exam-Style Questions

Q1 (lecture case 1, slide 30). A 53-year-old commercial fisherman has had fever, chills, malaise and **yellow-green productive cough for 1 day**. T 38.5, HR 96, RR 26, BP 98/62. **↑ fremitus and bronchial breath sounds**; CXR shows **right upper lobe consolidation**. He takes metoprolol and warfarin. Most likely pathogen? a. M. tuberculosis · b. Pseudomonas · c. Fusobacterium · d. Mycoplasma · e. Legionella · f. S. aureus · g. S. pyogenes · h. H. influenzae · i. Histoplasma · **j. S. pneumoniae**

Answer: j. S. pneumoniae. *[reasoning added; key not on slides]*

- **Acute (1 day)**, rigors, purulent sputum and **lobar consolidation** = typical CAP; pneumococcus causes 60%.
- Metoprolol explains the modest pulse.
- Distractors:
  - **Legionella** would bring diarrhoea, confusion and a water exposure.
  - **Mycoplasma** is insidious with a dry cough.
  - **Histoplasma** and **TB** are subacute or chronic.
  - **Pseudomonas** is a hospital/structural-lung organism.
  - **Fusobacterium** follows aspiration.

Q2 (lecture case 2, slide 31). A 67-year-old woman (HTN, DM, ex-smoker) has 2 days of fever, chest pain and purulent cough. T 39, HR 110, **RR 33**, **BP 143/88**, SpO_2 94%, crackles at the right upper zone, WBC 12 300, ESR 60, **BUN 15 mg/dL**, alert. Next step? a. Outpatient moxifloxacin · b. Inpatient cefepime + azithromycin + gentamicin · c. ICU, ceftriaxone + azithromycin · d. Outpatient azithromycin + amox-clav · **e. Inpatient azithromycin + cefotaxime** · f. ICU, amp-sulbactam + levofloxacin · g. Outpatient azithromycin · h. Inpatient ceftriaxone alone · i. ICU, aztreonam + levofloxacin

Answer: e. *[reasoning added]*

- **CURB-65 = 2**: age ≥ 65 (1) + RR ≥ 30 (1). She is not confused, BUN 15 is not > 19, and BP is normal.
- Score 2 → **hospitalisation** (slide 22), not ICU (that is ≥ 3).
- Inpatient regimen (slide 23) = **3rd-generation cephalosporin + azithromycin** (cefotaxime ≈ ceftriaxone) or a respiratory FQ.
- b is an MDR/HAP regimen. h misses atypical cover.

Q3. A patient has high fever, a pulse of 78, 10 days of dry cough, headache, and patchy interstitial infiltrates. Gram stain shows no organisms. Best empiric agent? A. Amoxicillin · B. **Azithromycin** · C. Ceftriaxone · D. Vancomycin · E. Nafcillin

Answer: B.

- Atypical CAP (Mycoplasma): no cell wall, so **penicillin-resistant** and **Gram stain useless** (slide 10). Macrolide is preferred (slide 24).

Q4. Seven days into a hospital stay, after IV antibiotics for cellulitis, a patient develops fever, purulent sputum and a new infiltrate. Empiric therapy? A. Amoxicillin · B. Ceftriaxone + azithromycin · C. **Piperacillin-tazobactam + levofloxacin + vancomycin** · D. Oral doxycycline · E. Clindamycin alone

Answer: C.

- **HAP with MDR risk** (IV antibiotics in 90 days, stay ≥ 5 days) → anti-pseudomonal β-lactam + non-β-lactam + MRSA cover (slides 26–27).

Q5. A 40-year-old healthy outpatient with CAP, CURB-65 0, who took amoxicillin 6 weeks ago. Best regimen? A. Amoxicillin again · B. Azithromycin alone · C. **Levofloxacin (or amoxicillin + macrolide)** · D. Ceftriaxone IV · E. Vancomycin

Answer: C.

- **Antibiotics in the past 3 months** moves him to the comorbid branch: **respiratory FQ or amoxicillin + macrolide** (slide 23).

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8. Exam Pitfalls & Traps

- **CURB-65 urea cut-off:** the slide uses **BUN > 19 mg/dL (> 7 mmol/L)**. In case 2, BUN 15 scores **0**.
- **Mortality figures** on slide 22: **1.5% / 9.2% / 22%**. Memorise them as written.
- **Recent antibiotics (≤ 3 months)** count like comorbidity: choose the FQ or amoxicillin + macrolide branch.
- **Inpatient ceftriaxone alone** misses atypicals. The slide pairs it with **azithromycin**.
- **Pneumococcus is the commonest cause of CAP overall.** Mycoplasma is only the commonest **atypical** cause.
- **Legionella:** confusion + diarrhoea + hyponatraemia → **urinary antigen**. Don't rely on the Gram stain.
- **HCAP is struck through** on slide 7: the category is obsolete *[added]*.
- **HAP MDR risks (slide 26):** IV antibiotics in 90 days, **≥ 5 days in hospital**, **RRT or ARDS before VAP**, **septic shock at VAP**.
- **Two different anti-pseudomonal classes** in dual therapy (β-lactam + FQ or aminoglycoside). Two β-lactams is wrong.
- **Slide 29 summary says CAP IV antibiotic = amoxicillin.** That is UK-style trust guidance and differs from slide 23 (ceftriaxone + azithromycin for inpatients). For MCQs follow **slide 23**.
- **Symptoms can persist for 3–6 months** after treatment. Don't over-investigate slow recovery.